



Synthesis of eugenol ester-based sily-acrylate polymer and its antifouling properties

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Abstract Marine biofouling has caused huge economic and energy losses, and the application of marine antifouling coatings is one of the effective solutions to this problem. Sily-acrylate based self-polishing coatings are favored because of excellent antifouling performance. However, weak static antifouling performance restricts the development of sily-acrylate based antifouling coatings. To solve this problem, an eugenol ester-based sily-acrylate polymer (EMSPs) is synthesised, which showed both self-polishing and antifouling performance. The highest content of eugenol (EMSP-16) inhibits *Staphylococcus aureus* (*S. aureus*) and *Escherichia coli* (*E. coli*) up to 62.2% and 76.3%, respectively, and inhibits the adhesion of *Halimphora* sp. and *Nitzschia closterium* (*N. closterium*) up to 85.1% and 90.8%, respectively. Mussel adhesion experiments show that mussels tend to avoid coatings with high eugenol content. The eugenol ester-based sily-acrylate polymer provides a new way to develop new environmentally friendly marine antifouling coatings and alleviate heavy metal pollution in the marine environment.

Keywords Marine antifouling, Self-polishing coating, Eugenol, Silane acrylate

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Introduction

With the continuous development of ship manufacturing and various marine industries, the damage caused by marine fouling organisms attached to the surface of marine equipment is increasing.^{1–6} Attachment of microfouling organisms, such as bacteria and diatoms, and macrofouling organisms, such as mussels and barnacles, to hull surfaces can exacerbate hull corrosion and fuel consumption,^{7–10} which in turn generates a series of economic and environmental problems. Applying marine antifouling coatings is currently known to be a cost-effective way to solve the marine biofouling problem.^{11–15} Tin-free self-polishing antifouling coatings are the most widely used antifouling coatings which are mainly composed of acrylic or methacrylate hydrolyzable resins as the base material, zinc oxide, cuprous oxide, and auxiliary antifouling agents such as copper pyrithione and zinc pyrithione (ZnPT) as the antifouling fillers.¹⁶ The antifouling activity of commercially available self-polishing antifouling coatings is still mainly provided by metal-based antifouling agents. The surface of the coating is polished and detached from seawater due to hydrolysis of hydrolysable polymers, thus releasing the antifouling agent filler inside the coating, which provides an antifouling effect,¹⁷ while the resin itself does not have antifouling ability. However, the release of highly toxic antifouling agents in paints can cause adverse ecological impacts, e.g., copper compound-containing antifouling paints have been banned globally due to toxicity. Furthermore, the most common zinc-based fungicide, zinc pyrithione (ZnPT), has been found to be toxic to some marine organisms.^{18,19} Therefore, it is particularly important to develop new environmentally friendly antifouling polymers.

Sily-acrylate polymer self-polishing antifouling coatings as long-lasting marine antifouling coatings, compared with other self-polishing coatings with energy-

saving drag reduction performance, through the process of silane hydrolysis to release antifouling agent, hydrolysis products do not contain heavy metal ions on the marine environment without pollution.^{20–23} At the present time, sily-acrylate polymers still use Cu_2O as the main antifouling agent component, copper pyrithione (CuPT) and zinc pyrithione (Zineb) as the auxiliary organic antifouling agent component, which will cause great harm to the marine environment. Furthermore, the polarity of the silicone-oxygen bond contained in the sily-acrylate polymer is smaller than that of the tin-oxygen bond contained in the organotin, which makes it more difficult for silicone to be attacked by nucleophilic reagents than tin, and thus the self-polishing performance of sily-acrylate self-polishing coatings is poorer than that of organotin self-polishing coatings, which results in the fact that the sily-acrylate based self-polishing antifouling coatings are only applicable to fast-moving ships. Sily-acrylate resin will exhibit poor antifouling performance when the ship's driving speed is lowered or the ship is at anchor for a long period of time.^{24–26} At the same time, sily-acrylate polymer self-polishing antifouling coatings indicate poor resistance to the adhesion of microorganisms, such as bacteria. Once a biofilm forms on the surface of the coatings, it attracts many microorganisms.^{27–29} Enhancing the antifouling performance of sily-acrylate based self-polishing antifouling coatings is a hot issue in current research. For example, 2-methylene-1, 3-dioxane (MDO), tributylmethoxysilane methacrylate (TBSM), and methyl methacrylate (MMA) are used to synthesize degradable methylsilane acrylate copolymers that have degradable backbone and hydrolyzable side groups, and the rate of self-polishing of the coatings can be controlled by the ratio of the side groups of the methylsilane ester to the ester units in the backbone.³⁰ Lin et al. prepared silicone-based polybenzoxazines with phenolic hydroxyl groups on the bottom and hydrophobic siloxane groups on the surface to enhance substrate adhesion.³¹ In a previous study, a diblock copolymer of tert-butyl dimethylsilyl methacrylate (TBDMSiMA) and methyl methacrylate (MMA) was used to prepare antifouling coatings with self-polishing properties through a reversible addition-fragmentation chain transfer (RAFT) polymerization reaction, which had a controllable microstructure and provided better corrosion resistance and higher antifouling efficiency compared with conventional copolymer coatings.³²

Eugenol, a compound naturally occurring in clove oil, has been used in a wide range of food, flavor, pharmaceutical and medical applications.^{33–35} For example, Pérez et al. found that the incorporation of thymol, eugenol and guaiacol into marine coatings showed significant antifouling activity.³⁶ Sha et al. found the hydrolysis of phenolic ester group realizes the transformation of resin segment from hydrophobicity to hydrophilicity. The self-polishing process releases eugenol, which is natural and non-toxic into seawater. It is the first time when self-polishing coating

polymer achieves self-polishing by hydrolysis of phenolic ester groups and release of natural phenolic products.³⁷ Therefore, the introduction of eugenol into sily-acrylate polymer self-polishing antifouling coatings is expected to improve the antifouling properties of the coatings. Moreover, the incorporation of eugenol enhances the hydrolysis rate of the resin and improves the self-polishing properties of the resin.^{38,39} The incorporation of eugenol into polymers is favorable for the development of new environmentally friendly marine antifouling coatings.

In this work, a simple one-step radical polymerization method is used to synthesize eugenol ester-based sily-acrylate polymer (EMSP). EMSPs coatings also have good surface self-renewal properties during hydrolysis, which helps to maintain the smoothness of the coating surface to reduce the resistance of ship navigation and realize energy saving and emission reduction. The EMSPs coatings showed good antifouling properties in the anti-diatom adhesion experiments and anti-bacterial adhesion experiments. Furthermore, the EMSPs series coatings with stable hydrolysis rate can controllably release environmentally friendly eugenol and silanol, which shows a good application prospect in the field of marine antifouling. The addition of eugenol enables EMSPs coatings to maintain good antifouling properties under static conditions while having the good dynamic self-polishing ability of acrylic silane ester resins to maintain a smooth surface. This provides a worthy approach to improve the performance of antifouling resins by adding natural antifouling actives.

Experimental

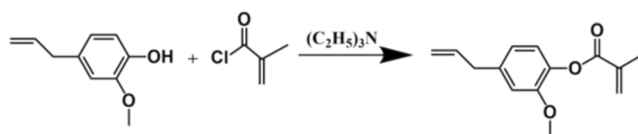
Materials

Eugenol (AR), methacryloyl chloride (AR) and triethylamine (AR) were purchased from Aladdin Company. Xylene, KBr (GR), ethyl acrylate (EA), methyl methacrylate (MMA), benzoyl peroxide (BPO) and azobisisobutyronitrile (AIBN) were purchased from Tianjin Guangfu Fine Chemical Research Institute (Tianjin, China). Triisopropylsilyl methacrylate (TISMA) was purchased from Inner Mongolia Sanyou Chemical Co., Ltd. Artificial seawater (ASW) used for diatom cultivation was prepared according to ASTM D1141-98, whereas the ASW in the film immersion experiment was prepared by reef salt provided by Seachem. All reagents were used without further purification unless otherwise specified.

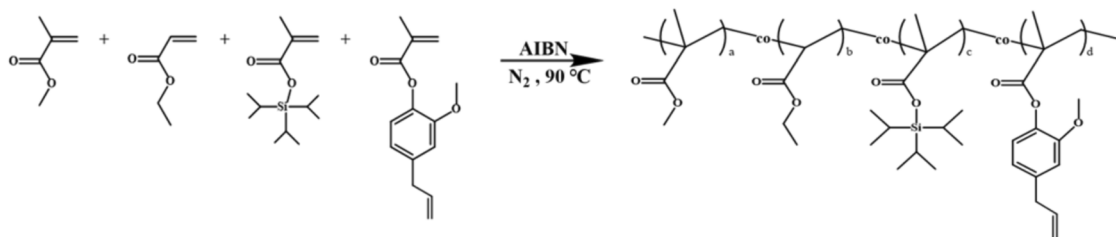
Synthesis of eugenol methacrylate and eugenol ester-based sily-acrylate polymer (EMSPs)

Eugenol methacrylate (EM) was prepared as shown in Scheme 1 (1). Water was removed from a quantity of

(1) Synthesis of eugenol methacrylate



(2) Eugenol methacrylate and eugenol ester-based sily-acrylate polymer



Scheme 1: Scheme for synthesis of the eugenol methacrylate and eugenol ester-based sily-acrylate polymer (EMSPs) via free radical polymerization.

xylylene using 3A molecular sieves, and a quantity of eugenol and triethylamine was taken and dissolved in the de-watered xylene at a molar ratio of 1:1.2. The mixture was stirred continuously in an iced salt bath until the temperature fell below 2°C. An equal molar amount of methacryloyl chloride was added slowly to the mixture. Stirring was continued in an ice-salt bath. A quantity of methacryloyl chloride (the molar amount of methacryloyl chloride is equal to the amount of eugenol) was added slowly dropwise to the flask from a constant pressure dropping funnel. After the dropwise addition, the device was kept sealed and stirred at room temperature for 24 h. The ammonium salt was removed by filtration, the unreacted monomer was removed by cationic adsorbent resin, and the solvent was evaporated under reduced pressure to obtain the monomer of eugenyl methacrylate.

EMSPs are prepared by free radical polymerization as shown in Scheme 1 (2). A quantity of xylene was added to a four-necked flask equipped with a thermometer and a spherical condenser tube and heated to 90°C. Subsequently, the mixture of MMA, EA, EM, TISMA, a chain transfer agent (MSD), and 3 g AIBN was added to the flask uniformly and slowly over a 3-h period. Upon completion, xylene (20 g) and BPO (0.75 g, 0.002 mol) were added to the flask in a homogeneous drop over a period of 30 min. The reaction was then continued with stirring for 2 h. Finally, the reaction was terminated by exposing the polymer to air (Table 1).

Characterization

The ^1H NMR spectroscopic measurements were acquired on a Bruker ARX 500 MHz spectrometer, using CDCl_3 as the deuterated solvent. The Fourier-transform infrared (FTIR) spectroscopy measurements were recorded on an AVATAR 360 FTIR spectropho-

Table 1: Formulation of the EMSPs

Sample	MMA/EA/EMA/TISMA (g)
EMSP-0	36/24/0/40
EMSP-4	33.6/22.4/4/40
EMSP-8	31.2/20.8/8/40
EMSP-12	28.8/19.2/12/40
EMSP-16	26.4/17.6/16/40

tometer in the 400–4000 cm^{-1} range by using the KBr-disk method. The scanning electron microscopy (SEM) images and element mappings were obtained from a field emission scanning electron microscope (SEM, Hitachi S4800). The X-ray photoelectron spectroscopy (XPS) measurements were performed using a PHI 5700 ESCA spectrometer with $\text{Al-K}\alpha$ X-ray source.

Results and discussion

Characterization of EM and EMSPs

The molecular structure of EM is investigated by ^1H NMR spectra and FTIR spectra. The results of ^1H NMR spectra are shown in Fig. 1a. The position of each peak corresponds to the following: ^1H NMR (500 MHz, CDCl_3 , ppm): 5.06–5.19 a, b ($\text{CH}_2 = \text{CHCH}_2-$), 5.9–6.1 c ($\text{CH}_2 = \text{CHCH}_2-$), 3.34–3.41 d ($\text{CH}_2 = \text{CHCH}_2-$), 6.80–6.75 e ($-\text{PhH}$), 6.97–7.00 f ($-\text{PhH}$), 6.82 g ($-\text{PhH}$), 3.823 h ($\text{CH}_3\text{O}-$), 6.66–6.72 i ($-\text{C}=\text{CH}_2$), 6.42–6.34 j ($-\text{C}=\text{CH}_2$), 2.13–2.02 k ($-\text{CH}_3$). As shown in Fig. 1b, the characteristic peak of EM at 1268 cm^{-1} is attributed to C–O–C stretching vibration peak, indicating the successful synthesis of EM via nucleophilic substitution reaction. The FTIR spectra of the EMSP polymers are shown in Fig. 1c. Compared

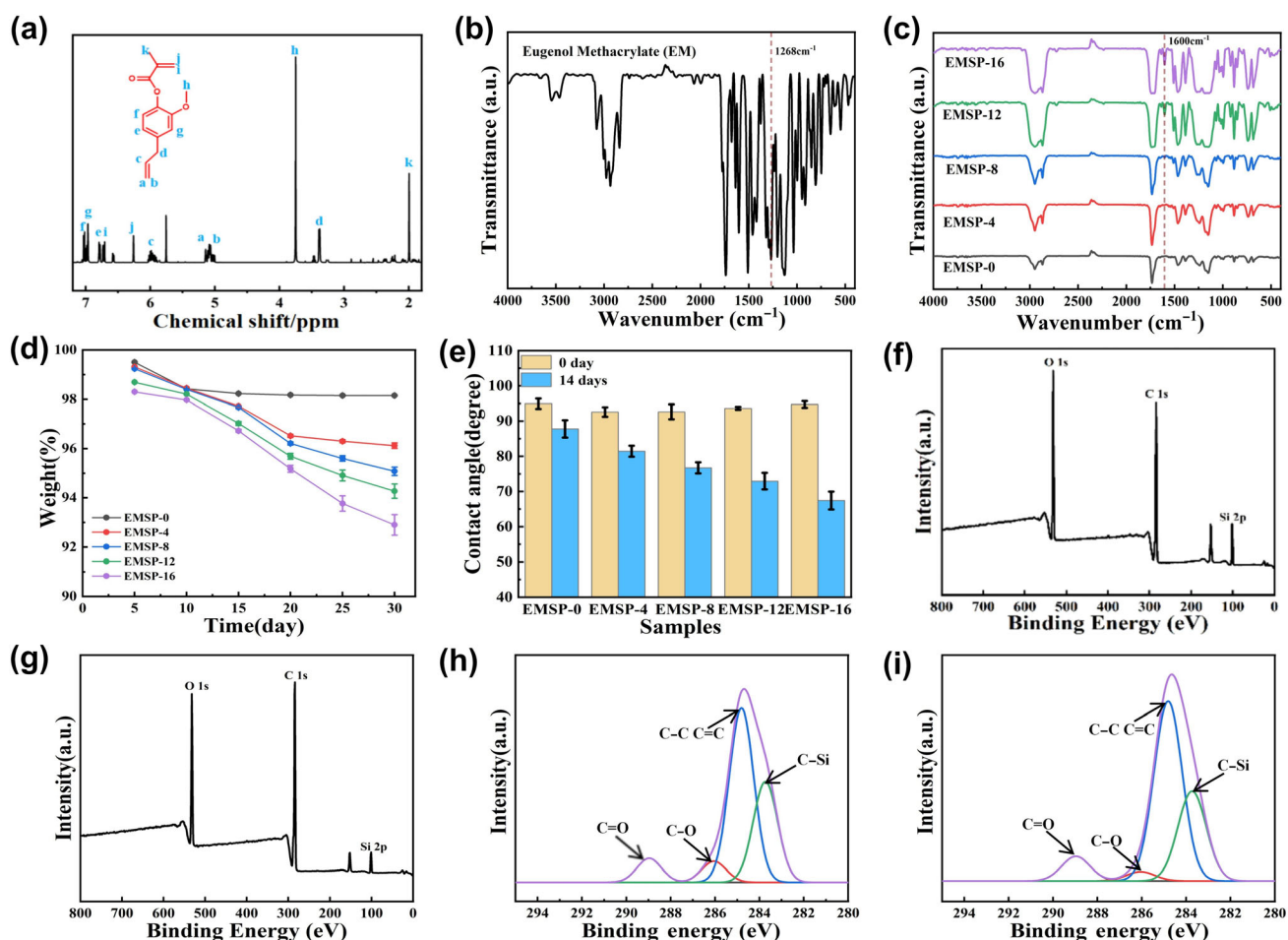


Fig. 1: (a) ^1H NMR spectra of EM. (b) FTIR spectra for EM. (c) FTIR spectra for EMSPs. (d) Weight loss ratios of EMSPs coatings. [number of repeat experiments] = 3 (e) The contact angle of different coatings before and after immersion in artificial seawater for 14 days. [number of repeat experiments] = 3 (f) XPS image of EMSP-16 before immersion. (g) XPS image of EMSP-16 after 30 days of immersion in artificial seawater. (h) Composition of C1s on the surface of EMSP-16 before immersion. (i) Composition of C1s on the surface of EMSP-16 after 30 days of immersion in artificial seawater

with the EMSP-0 polymer without eugenol methacrylate, other EMSPs polymers show a telescopic vibrational peak of the benzene rings at 1600 cm^{-1} , and the peak intensity increases with the increasing of the content of eugenol, confirming that the EMSPs polymers are successfully prepared.

The weight loss rate is an important parameter to measure coatings hydrolysis and antifouling performance. In this experiment, the hydrolysis rate of the coating is reflected by measuring the weight loss rate of the coatings during 30 days of immersion in artificial seawater. As shown in Fig. 1d, the EMSP-0 coating without the addition of EM monomers exhibits a significant weight loss in the first 10 days, but the mass change is not significant in the following 20 days. As the amount of EM monomers increases, the self-polishing performance of the coatings increases significantly. The more content of EM, the faster the weight loss of the coatings. The weight loss of EMSP-16 is the highest after 30 days, reaching 7.1%, which indicates

that the incorporation of EM monomers increase the hydrolysis capacity of the coatings.

Surface wettability exhibits a significant impact on the overall performance of self-polishing antifouling coatings and the water contact angle is measured to assess hydrophilicity. If the coating is too hydrophilic, it will accelerate the rate of polishing and reduce service life. As shown in Fig. 1e, there is little difference in the contact angle of the coatings before immersion. After 14 days of immersion, the higher the EM content in the EMSPs coatings, the greater the hydrophilicity of the coatings and the smaller the contact angle. This is partly due to the fact that the more phenolic ester groups on the surface of the coating, the faster the rate of hydrolysis, and the carboxyl groups exposed after hydrolysis are hydrophilic functional groups and the increase of hydrophilic functional groups makes the water contact angle decrease, easy to fall off from the surface of the coating to accelerate hydrolysis.^{40–42} On the other

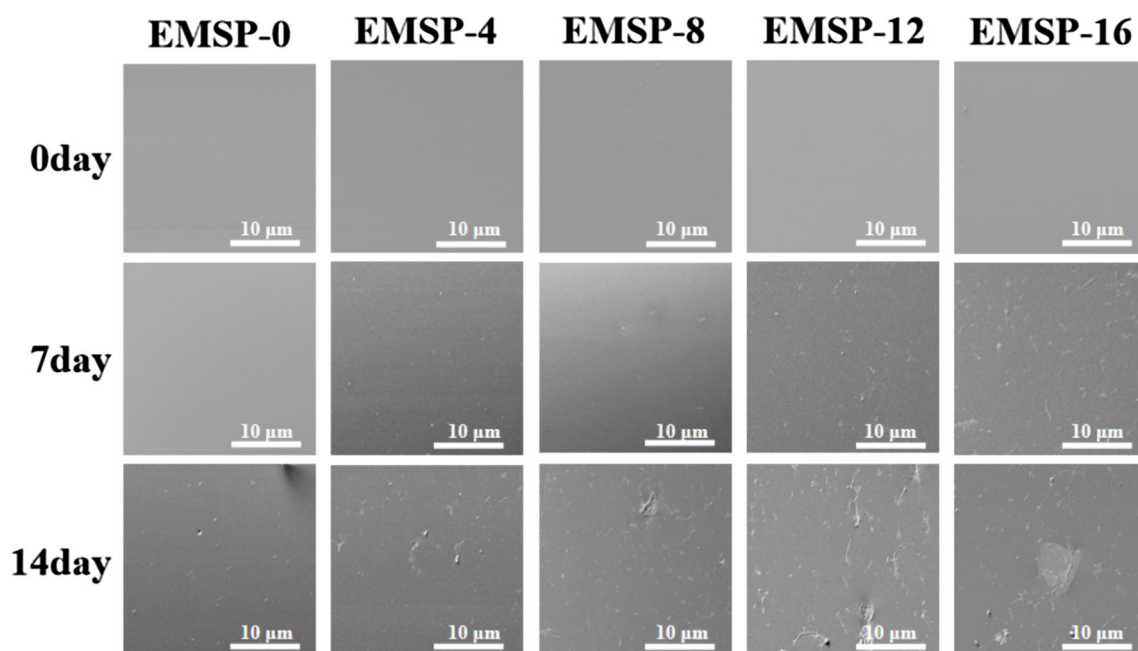


Fig. 2: Representative SEM images of EMSPs coatings before immersion and after 14 days immersion in artificial seawater

hand, silyl-acrylate resin itself is hydrophobic but its slow hydrolysis will also make the coatings to hydrophilic. Comprehensive weight loss curve and water contact angle test results show that the hydrolysis rate of EMSPs coatings is stable, and it still has good antifouling ability under static conditions, which is conducive to the realization of long-lasting antifouling. In order to study the hydrolysis of EMSP coatings, the coatings are immersed in artificial seawater for 30 days, and then immersed in deionized water for 2 h to remove the salts on the coatings, and then XPS test was conducted after drying (Figs. 1f–1i). By comparing Figs. 1f and 1g it can be found that before and after coating immersion the C-O and C-Si bond strengths decreased at 286.05 eV and 283.73 eV, respectively, indicating that the resin underwent a hydrolysis reaction during immersion, which led to the breakage of the side-chain C-O bonds. As can be seen in Figs. 1h and 1i, the percentages of Si and O elements on the surface of the coatings before and after immersion decreased significantly, which indicate the release of silanol groups into seawater.

Fouling organisms are less likely to adhere to flat coating surfaces because rough surfaces result in an increased contact area between the coating and seawater, thus providing more adhesion sites for fouling organisms. Therefore, the stabilization of the surface morphology of the coatings before and after immersion in seawater is important for the antifouling performance of the coating. Scanning electron microscopy (SEM) is used to investigate the surface morphology of EMSPs coatings before and after immersion in artificial seawater for 14 days, and the results are shown in Fig. 2. The surfaces of the coatings

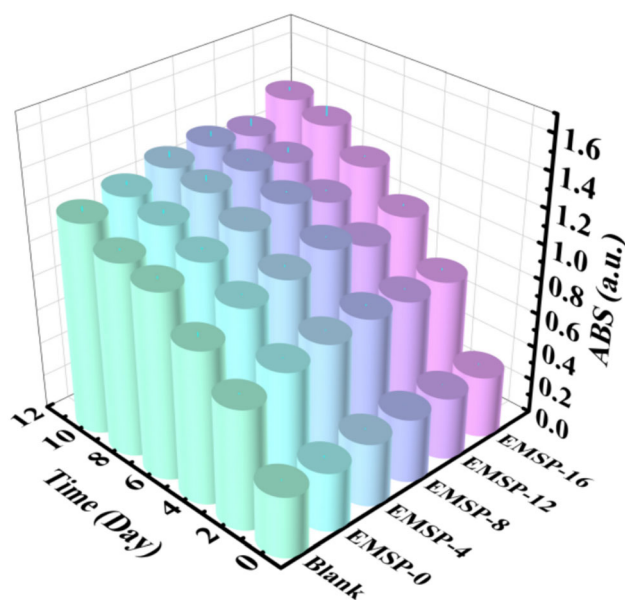


Fig. 3: Concentration changes of *Nitzschia closterium* after putting in different coatings

before immersion are relatively flat and smooth without cracks or wrinkles, which indicates that the EMSP coatings have good film-forming properties. The roughness of the coatings surface increases slightly after 14 days of immersion in artificial seawater, and the bumps of the coating surface also increase with the rise of EM content. However, the overall coating remains flat, which indicates that the coating is hydrolyzed uniformly.

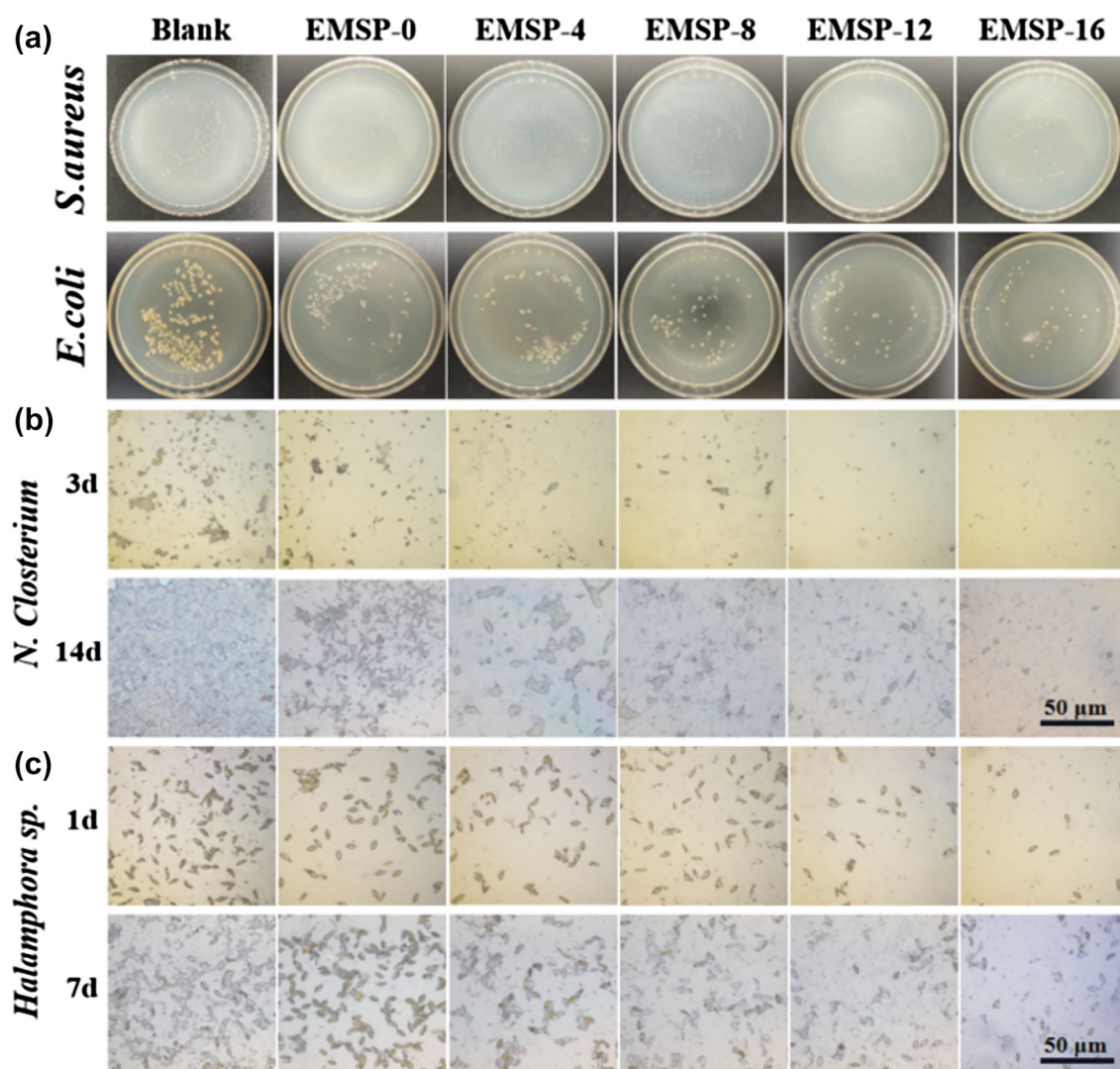


Fig. 4: (a) The digital photographs of the colonies incubated on agar plates with different EMSP coatings toward *E. coli* and *S. aureus* (b) Representative microscopy images of different EMSP coatings immersed in *N. closterium* after 3 days and 14 days. (c) Representative microscopy images of different EMSPS coatings immersed in *Halamphora sp.* after 1 day and 7 days. [number of repeat experiments] = 3

The toxicity of antifouling coatings can adversely affect marine ecology, thus environmental friendliness is an important indicator of their performance. In order to evaluate the toxicity of EMSP coatings, EMSPs are coated on the surface of slides and immersed in the *N. closterium* solution for 10 days, and the concentration of the algal liquid is measured by UV spectrophotometer to study the effect of EMSP coatings on the growth rate of *N. closterium*. As shown in Fig. 3, the algal concentration of *N. closterium* gradually increases. There is no significant difference between the algal solutions immersed with different EMSP coatings, indicating that the hydrolysis of the EMSP coatings does not hinder the normal growth of algae and it is a kind of environmentally friendly coating.

Antifouling properties

To evaluate the antifouling performance of EMSPs coatings, the inhibition of bacterial adhesion on the coating surface is performed. *E. coli* and *S. aureus* are used as models to test the bacterial adhesion inhibition performance of EMSPs coatings. As shown in Fig. 4a, the EMSPs coatings show significant adhesion inhibition performance against *S. aureus* and *E. coli*. The inhibition rates of EMSPs against *S. aureus* are obtained by calculating the number of bacterial communities on the plate and controlling with the blank samples as 13.3%, 39.0%, 49.4%, 55.3% and 62.2%, respectively. It can be seen that antimicrobial activity of *S. aureus* increases significantly with the increase of EM content. In contrast, EMSP-0, EMSP-4, EMSP-8, EMSP-12 and EMSP-16 inhibit *E. coli* by 16.1%,

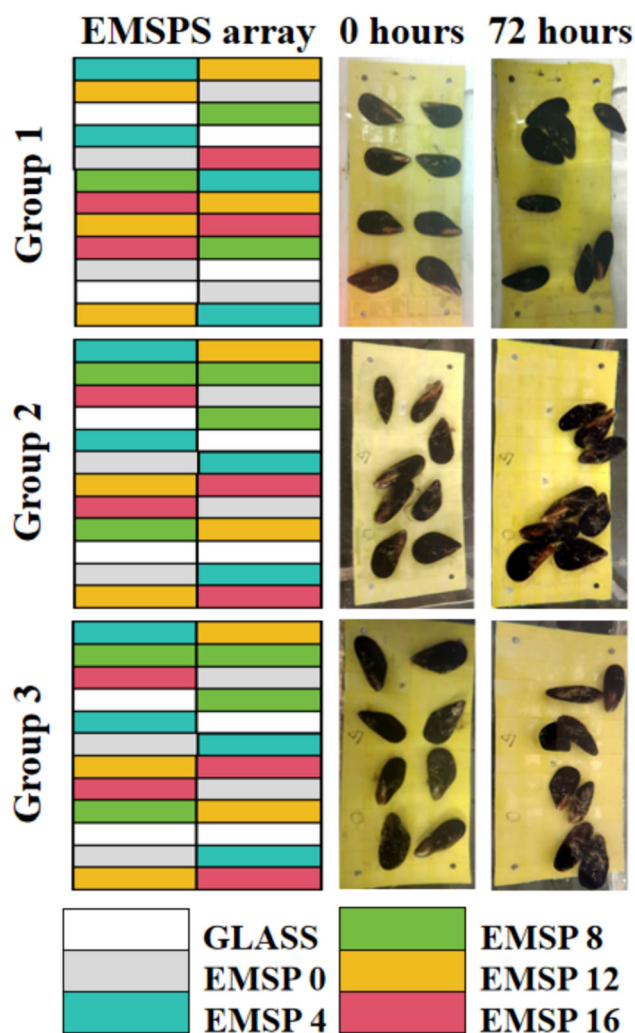


Fig. 5: Mussels settlement and choice assays: positions of the mussels at the beginning and after 72 h. [number of repeat experiments] = 3

49.4%, 62.6%, 71.7% and 76.3%, respectively, which is slightly higher than that of *S. aureus* overall. It can be observed that the bacterial adhesion inhibition performance is significantly enhanced with the increase of EM content in the EMSPs coating. The results indicate that the EM content contained in the EMSPs coating can effectively inhibit bacterial adhesion.

N. closterium and *Halamphora sp.* were selected to evaluate the anti-diatom properties of EMSP coatings. The results show in Figs. 4b and 4c that the EMSPs coatings indicated significant anti-adhesion properties against *N. closterium* and *Halamphora sp.*, and the anti-adhesion properties were enhanced with the mass fraction of EM monomer. After 7 days, EMSP-0, EMSP-4, EMSP-8, EMSP-12 and EMSP-16 inhibited the attachment rate of *Halamphora sp.* by 20.9%, 43.0%, 50.2%, 65.8% and 85.1%, respectively. After 14 days, EMSP-0, EMSP-4, EMSP-8, EMSP-12 and EMSP-16 inhibited the attachment rate of *N. closterium* by 46.5%, 65.4%, 75.4%, 84.3%, and 90.77%,

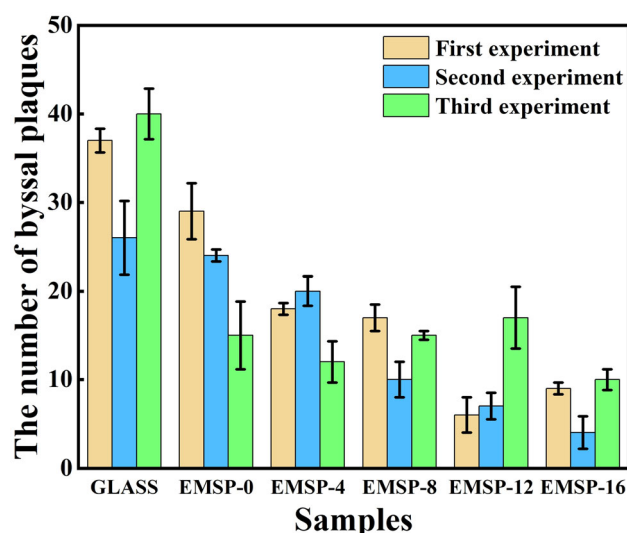


Fig. 6: The average number of byssal plaques adhered to different EMSP coatings. [number of repeat experiments] = 3

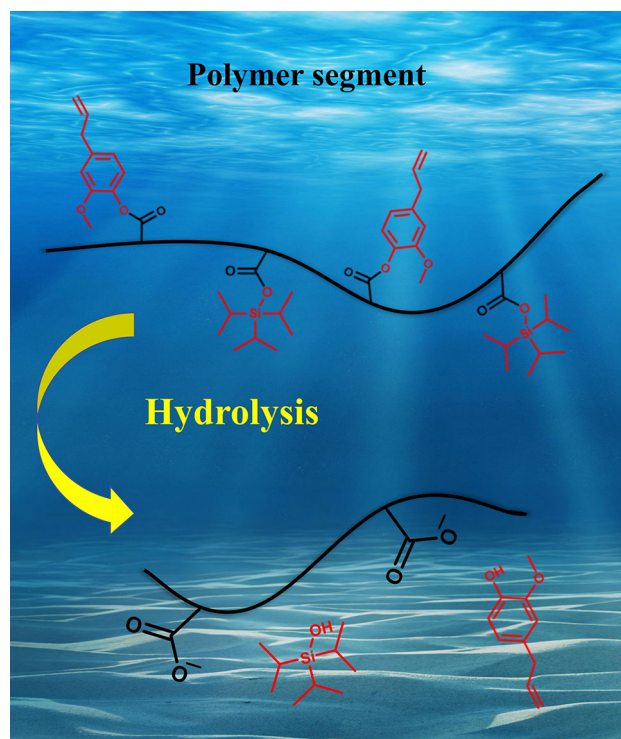


Fig. 7: Hydrolysis schematic illustration of the EMSPs

respectively. The increase of EM content in the coatings improved the anti-algal adhesion performance, and the inhibition ratio of the coating on the *N. closterium* was higher than that of the more strongly adhering *Halamphora sp.* Good anti-diatom performance indicates that EMSPs coatings have a broad application prospect in the actual marine antifouling.

It is well known that mussels are typical marine fouling organisms that are large in size, mass and highly hazardous.^{43–46} Under laboratory conditions, mussels, a common fouling organism, are selected to investigate the anti-mussel adhesion of EMSPs. At the beginning of the test, the EMSPs series coatings are randomly applied to a plate using glass slides as substrate and the mussels are evenly placed on the test plate. Mussel movement trajectories and biological activities are recorded by digital photography. To comprehensively assess the effect of the coatings on the selective attachment behavior of the mussels, three parallel replicated experiments are conducted. Figure 5 shows the results of the digital photographs taken at the beginning of the experiment and 72 h later. Figure 6 shows the number of byssal plaques adhered to different coating surfaces in the three parallel experiments. In the first group, five mussels produce movement, three of them report at the junction of EMSP-12, glass slides and EMSP-0, and they produce 19 byssal plaques attached to the glass sheet. In contrast, only two and three are attached to EMSP-12 and EMSP-0, respectively, and the other two tend to leave the test area from EMSP-0 and move from EMSP-16 to embrace with another mussel on the glass slides, respectively. In the second group, three mussels move from EMSP-16 and EMSP-8 to the area of the glass slides, EMSP-8 and EMSP-0 to clump and most of the filaments are attached to the glass slides, and five mussels show clumping behavior with most of their filaments attached to the junction of the glass sheet. In the third group, four mussels show attachment behavior to the glass slides, two to EMSP-0 and two to EMSP-4. Overall, most of the mussels show a tendency to move towards or away from the test area in the area of the samples with low EM content with the highest number of attachments in the area of the glass slice and EMSP-0, and almost no attachment in the area of the samples with higher EM content, proving that the increase of the EM content in the EMSP improves the mussel resistance of the coating.

Hydrolysis mechanism of the EMSPs

Antifouling illustration of EMSPs is showed in Fig. 7. Compared with copper/zinc acrylic antifouling coatings, sily-acrylate polymer self-polishing antifouling coatings react only with water during hydrolysis, and their reaction products do not contain heavy metal ions.⁴⁷ Sily-acrylate polymer self-polishing antifouling coatings are continuously hydrolyzed in seawater and peeled off layer by layer to inhibit the adhesion of fouling organisms.^{48,49} Moreover, EM monomer is introduced into the sily-acrylate system to enhance the self-polishing ability of the resin. By hydrolyzing the resin in weakly alkaline seawater (pH = 8.3), eugenol is released from the resin (Fig. 7) and inhibit the adhesion and further growth of marine primary fouling organisms.^{50,51}

Conclusions

In summary, a novel self-polishing antifouling coating polymer is prepared, which exhibits excellent environmental friendliness, controllable self-polishing properties, and static antifouling ability compared to conventional marine antifouling coatings. The self-polishing EMSPs coating inhibits the adhesion of fouling organisms by releasing eugenol through the hydrolysis sequence. Antibacterial tests conducted under laboratory conditions show that EMSP-16 inhibits *S. aureus* and *E. coli* by 62.2% and 76.3%, respectively. The anti-diatom adhesion test shows that EMSP-16 inhibits the adhesion of *Halamphora sp.* and *N. closterium* by 85.1% and 90.8%, respectively. The above experiments show that the polymer has good anti-algal and antibacterial properties. Mussel adhesion experiments show that mussels tend to avoid coatings with high eugenol content. The eugenol ester-based sily-acrylate polymer provides a new way to develop new environmentally friendly marine antifouling coatings and alleviate heavy metal pollution in the marine environment.

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Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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